Electric drives

Speed controlling + Energy transferring / conversion

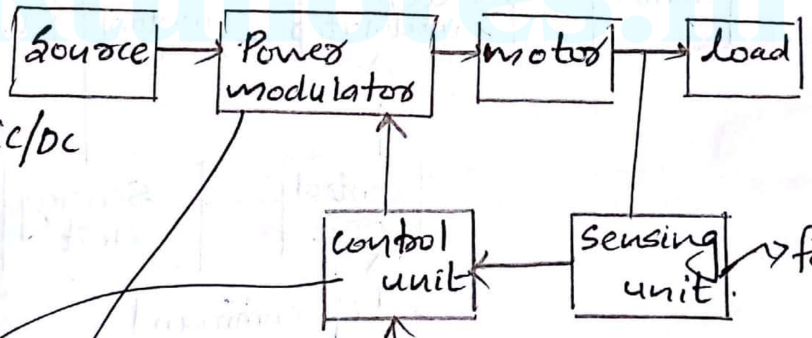
eg: electric motor
fan (TRAIC, AC motor)

electric motor + speed/movement control

→ Induction motor (main AC)
Sync. motor
DC. motor.

domestic and industrial applications.

Parts of electric drives.



AC/DC

→ for protecting the ckt from fault.

Give pulses to the ckt
IC's, μp, μc.

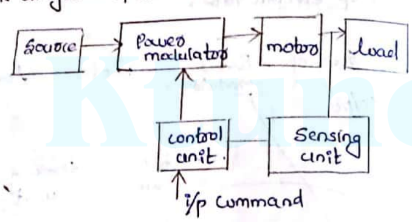
→ transferring power from source to load.
Protecting device.
Switching ckt or variable impedence ckt
power electronic ckt.

AC→DC (Rectifier), DC→AC (Inverter), DC→DC (chopper),
AC→AC (AC voltage regulator)(TRIAC)

Electric drives

Motion control is required in large number of industrial and domestic applications like transportation s/ms, washing machines, rolling mills, textile mills, fans, pumps etc. Systems employed for motion control are called drives and may use any of the prime movers such as diesel or petrol engines, gas or steam turbines, electric motors for supplying mechanical energy for motion control. Drives employing electric motors are known as electric drives.

Block diagram of an electric drives



Load is usually a machinery designed to accomplish a given task.
example : fans, pumps, washing machine, trains etc
usually load requirements can be specified in terms of speed and load demands. A motor having same speed-torque characteristics as that of the load requirements is choosen.

Power modulator

It performs following functions.

1. Modulates the flow of power from the source to the motor.
2. Converts electrical energy of the source in the form suitable to the motor then it is appropriately called converter
3. During transient operation such as starting, braking and speed reversal. It restricts the source and motor current within the permissible value. Also excessive current drawn from the source may overload the motor or it may cause a voltage dip.
4. Select the mode of operation of the motor that is motoring or braking (Regenerating action)

Controls for power modulator are built in control unit which usually operates at much lower voltage and power levels. in addition to operating the power modulator as desired. It also generate commands for the protection of power modulator and motor. Input command signal forms an input to the control circuit unit. It also adjust the operating point of the drive. Sensing of certain drive parameters such as motor current and speed is required either for protection or for closed loop operation.

Applications of electricdrives

Dc motor drives

- Paper, sheet, film winding and unwinding machines
- Extruder screws
- Lathes, milling machines, boring machines, spindless and feeds of machine tools.
- Roller tables in mills
- Rubber mixers
- Motor braking systems.

Sync motor drives

- Useful in applications requiring precise speed and position control.
- Speed and position may be accurately controlled using openloop controls. eg: stepper motors.
- low power applications include positioning machines, where high precision is required & robot actuators
- Record players turntables.
- Ball mills - Increased efficiency in low speed applications

Induction motor drives

- Traction motors
- Simplification and analysis of equivalent ckt.
- Refrigerators, washing machines, & furnaces as well as conveyors, pumps, winders, wind tunnels and other industrial equipment.

Parts of electrical drive

1. load
2. motor
3. Powermodulator
4. Control unit
5. Sources
6. Sensing unit

1. Sources

Very low power drives are fed from single phase source rest of the drives are fed from 3ø source. most of the drives are powered from AC source either directly or through a converter link. when fed directly from 50Hz AC supply, maximum speed of AC motors are limited to 3000 rpm. for higher speed, conversion to high efficiency supply is required. In the case of aircraft and space applications, 400Hz AC supply is used. for main line traction high voltage supply is preferred. In India 25kV, 50Hz supply used. Some drives are powered from battery.

2. Powermodulator

They are classified as

a. converters
b. Switching ckts.

c. variable impedences.

Power modulators are the devices which alter the nature of voltage or frequency as well as changes the intensity of power from the source to control an electrical drive.

a) Converters

Depending upon the type of functions converters can be divided into 5 types

1. Ac to Dc converter

Fixed Ac → controlled rectifier → Variable dc

2. Ac regulators (Ac to Ac converters)

Fixed Ac → Ac regulator → Variable AC

3. Dc to Dc converter (chopper)

Fixed DC → chopper → Variable DC

4. Dc to Ac converter

Fixed DC → Inverter → Variable AC

5. Cycloconverters.

It converts the i/p power at one frequency

Fixed voltage, Fixed frequency AC → Cycloconverter → Variable voltage, Variable frequency AC

b) Switching ckts

Switching ckts in motors and electrical drive are used for running the motor smoothly and also protects the machine during faults. These ckts are used for changing the quadrant of operation during the running condition of the motor. These ckts are implimented to disconnect the motor from the main ckt during any abnormal conditions.

c) Variable impedences

These are used for controlling the speed by varying the resistance or impedence of the ckt. But these control methods are used only in low cost dc and AC drives. Here the resistance can be varied in steps by means of contactors.

3. Electric motors

Most commonly used electric motors in an electrical drives are dc motors, induction motors, sync motors, stepper (pointer) motors, switched reluctance motors etc.

Due to the development of semiconductor converters AC motors are now used in variable speed drives. due to the presence of commutator and brushes dc motor have a number of disadvantages as compared to AC motors. Also, dc motor is having higher cost, more weight and volume. due to all these disadvantages AC drives are used in variable speed applications.

4. Control unit

Controls for a power modulator are provided in the control ckt. Nature of the control unit for a particular drive depends on the power modulator used. when Semiconductor converters are used, the control unit will consists of firing ckt which employ linear and digital IC's, transistors and a μp.
when control of switching ckt is required the function of control unit will be sequencing and interlocking.

5. Sensing unit

It senses the motor parameters by means of various sensors to provide protection during abnormal condition and feed back the sensed value to the control circuit.

Choice of electrical drive (factors when we have to choose an electric drive for a particular application.)

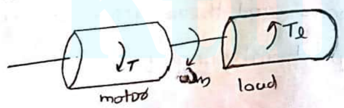
factors affecting the choice of electrical drives are

1. Steady state requirements
Nature of speed - torque characteristics, speed regulation, speed range, efficiency, duty cycle, quadrants of operation etc.
2. Transient state operation requirements.
Values of acceleration and deceleration starting, breaking and reversing performance
3. Requirements related to the Source
Type of source and its capacity, magnitude of voltage, Voltage fluctuations, PF, harmonics and there affect on other loads
4. Capital and running cost, maintenance needs, life
5. Environment and location
6. Reliability.
7. Space and weight
8. Service requirements.

Dynamics of electrical drive.

Fundamental torque equation of an electrical drive S/m

A motor generally drives a load (machine) through some transmission S/m. While the motor always rotates, the load may rotate or may undergo translational motion. Load speed may be different from that of motor, and if the load has many parts, there speeds may be different and while some may rotate, others may go through a translational motion. Therefore a motor load S/m is represented by an equivalent rotational S/m.



[change in motion]

Let J – moment of inertia of motor load S/m referred to the motor shaft in kgm².

ω_m – Instantaneous angular velocity of motor shaft in rad/sec.

T – Instantaneous value of developed motor torque in N-m.

T_l – Instantaneous value of load torque referred to the motor shaft.

We have torque = moment of inertia × angular acceleration.

∴ Here in motorload S/m net torque,

$$T - T_l = J \times \frac{d\omega_m}{dt}.$$

$$\boxed{T = T_l + J\frac{d\omega_m}{dt}.}$$

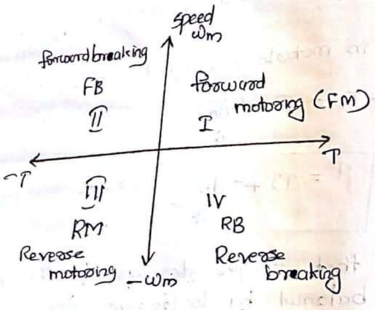
It shows that torque developed by the motor is counter balanced by load torque T_l and dynamic torque $J\frac{d\omega_m}{dt}$.

Dynamic torque is present only during transient operations.

During acceleration, motor should supply not only load torque but an additional torque $J\frac{d\omega_m}{dt}$ inorder to overcome the drive inertia.

During Decceleration, dynamic torque $J\frac{d\omega_m}{dt}$ has a negative sign. Therefore it asist the motor developed torque T.

4 Quadrant operation of an electric drive (multiquadrant)



Speed – torque conversions and 4th quadrants of operation of an electrical drive are explained as follows.

Motor speed is considered positively when rotor in the forward direction. For drives which operate only in one direction, forward speed will be there normal speed.

A motor can operate in 2 modes.

1) Motoring and braking

In motoring, It converts electrical energy into mechanical energy, It which supports its motion.

In braking it works as a generator converting ME into EE and thus opposes the motion.

Motor can provide motoring and braking operations for both forward and reverse directions.

Power developed by the motor is given by

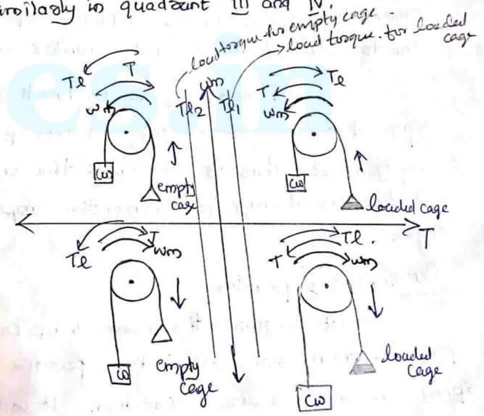
$P = T \times \omega_m$

or $T \rightarrow$ motor torque

In quadrant I, developed power is positive and speed ω_m is positive hence It is forward motoring mode.

In quadrant II power is negative, speed is positive $\therefore$ It is forward braking mode.

Similarly in quadrant III and IV.



Let us consider the operation of a hoist in 4 quadrants shown in fig above. Direction of motor torque, load torque and direction of speeds are marked by arrows.

A hoist consists of a rope wound on a drum coupled to a motor shaft. One end of the rope is tied to a cage which is used for transporting the material. One end of the rope has the counter weight. weight of the counterweight is choosen higher than weight of the empty cage, but lower than a fully loaded cage.

load torque T_{l1} in quadrants I and IV represents the speed-torque chara of loaded cage. This torque is the difference of torque due to loaded hoist and counter weight.

load torque T_{l2} in quadrants II and III represent the speed-torque chara of an empty hoist. This is -ve bcz the counter weight is always higher than the empty cage.

Quadrant one operation

Hoist requires the movement of cage upwards, which corresponds to the positive motor speed in anticlockwise direction. It will be obtained if the motor produce positive torque in anticlockwise direction. Since, the developed power is positive, this is forward motoring operation.

Quadrant II operation

It is obtained when an empty cage is moved up. since the counter weight is higher than the empty cage, It is able to pull it up. Inorder to limit the speed to a safer value, ~~motor~~ motor must produce a breaking torque in clockwise direction, since speed is positive, developed power is negative. It is forward breaking operation.

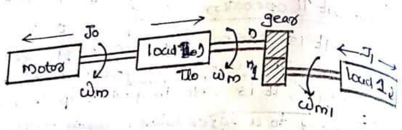
Quadrant 3 operation

Empty cage is lowered since empty cage wait is lessthan the counter weight. Thus motor produces a torque in clockwise direction. the speed is negative and developed power is positive. It is Reverse op motoring operation

Quadrant 4 operation

It is obtained when a loaded cage is lowered, Since the wait of loaded cage is higher than the counter weight, the loaded cage is moves in downward direction. Inorder to limit the speed of the cage within a safer value, motor must produce a positive torque in anticlockwise direction. both power and speed are negative It is reverse breaking operation.

Loads with rotational motion



Let us consider a motor driving two loads, one coupled directly to its shaft and other through a gear with n and n_1 teeth shown in the figure above. Let the moment of inertia of motor and load directly coupled to its shaft be J_0, motor speed and torque of the directly coupled load be ω_m and T_{l0} respectively. Let the moment of inertia, speed and torque of the load coupled through a gear be J_1, ω_{m1} and T_{l1} respectively.

Now geartooth ratio a_1 be

$$\boxed{a_1 = \frac{\omega_{m1}}{\omega_m} = \frac{n}{n_1}} \quad \text{------(1)}$$

If the losses in the transmission are neglected then kinetic energy due to equivalent inertia must be same as the sum of kinetic energy of various moving parts. Thus

$$\tfrac{1}{2} J\omega_m^2 = \tfrac{1}{2} J_0\omega_m^2 + \tfrac{1}{2} J_1\omega_{m1}^2 \quad \text{------(2)}$$

dividing

from eqn (1) $\omega_{m1} = a_1\omega_m$

Substituting in eqn (2),

$$\tfrac{1}{2} J\omega_m^2 = \tfrac{1}{2} J_0\omega_m^2 + \tfrac{1}{2} J_1(a\omega_m)^2$$

Dividing throughout by $\tfrac{1}{2}\omega_m^2$,

$$\boxed{J = J_0 + J_1 a_1^2}$$

If n other loads with moment of inertia $J_1, J_2, \cdots J_n$ and geartooth ratios of $a_1, a_2 \cdots\cdots a_n$ are connected in addition to a load which is directly coupled to the motor then the equivalent moment of inertia

$$\boxed{J = J_0 + J_1 a_1^2 + a_2^2 J_2 + a_3^2 J_3 + \cdots a_n^2 J_n^2}$$

Power at the loads and the motor must be same. T_l is the total equivalent load torque referred to the motor shaft. We have Power $= T \times \omega$ therefore

$$T_l \times \omega_m = T_{l0} \times \omega_m + T_{l1}\omega_{m1}$$

$$= T_{l0}\omega_m + T_{l1} a_1\omega_m$$

$$\boxed{T_l = T_{l0} + a_1 T_{l1}}$$ (for geartransmission efficiency $\eta = 100\%$).

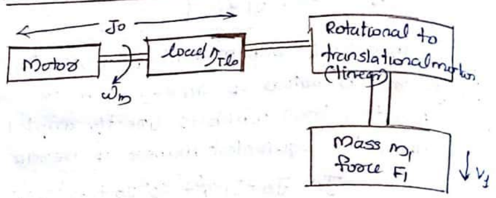
Generally if the geartransmission efficiency η then the equivalent load torque of the s/m

$$T_l = T_{l0} + \frac{a_1 T_{l1}}{\eta_1}$$

If n other loads are connected through gears having gear tooth ratios $a_1, a_2, \cdots\cdots a_n$ and gear transmission efficiencies $\eta_1, \eta_2, \ldots \eta_n$ then

$$\boxed{T_l = T_{l0} + \frac{a_1 T_{l1}}{\eta_1} + \frac{a_2 T_{l2}}{\eta_2} + \cdots + \frac{a_n T_{ln}}{\eta_n}}$$

<u>Loads with translational motion</u>



Let us consider a motor driving 2 loads, one coupled directly to its shaft and other through a transmission s/m. converting rotational motion to linear motion.

Let J_0 - moment of inertia of motor and load directly coupled to its shaft.

T_{l0} - Load torque of the ~~motor~~ load which is directly coupled to the motor shaft.

M_1 - mass in kg.

V_1 - linear velocity in m/s.

F_1 - force on the load VF_1 in Newtons

If the transmission losses are neglected then KE due to equalent inertia J must be same as the KE of various moving parts. That is,

$$\frac{1}{2} J\omega_m^2 = \frac{1}{2} J_0\omega_m^2 + \frac{1}{2} M_1 V_1^2.$$

dividing throughout by $\frac{1}{2}\omega_m^2$

$$J = J_0 + M_1\left(\frac{V_1}{\omega_m}\right)^2.$$

If n other loads are connected to the motor shaft through rotation to linear transmission s/m then equalent inertia J =

$$\boxed{J = J_0 + M_1\left(\frac{V_1}{\omega_m}\right)^2 + M_2\left(\frac{V_2}{\omega_m}\right)^2 + \cdots M_n\left(\frac{V_n}{\omega_m}\right)^2}$$

Similarly, power at the motor and the load should be same and if transmission efficiency ~~s/m~~ be η

$$T_l \omega_m = T_{l0}\omega_m + \frac{F_1 V_1}{\eta_1}$$

$\div \omega_m$, equalent load torque,

$$T_l = T_{l0} + \frac{F_1 V_1}{\omega_m \eta_1}$$

If n other loads are connected to the transmission s/m having linear motion

$$\boxed{T_l = T_{l0} + \frac{F_1 V_1}{\eta_1 \omega_m} + \frac{F_2 V_2}{\eta_2 \omega_m} + \cdots \frac{F_n V_n}{\eta_n \omega_m}}$$

? A motor drives 2 loads. One has rotational motion and it is couple to the motor through a reduction gear with $a = 0.1$ and η of 90%.

The load has a moment of inertia of 10 $kg\,m^2$ and a torque of 10Nm. other load has translational motion and consists of 1000kg weight to be lifted at an uniform speed of 1.5 m/s. coupling b/w the load and the motor has an η of 85%. motor has an inertia of 0.2 $kg\,m^2$ and runs at a constant speed of 1420 rpm. Determine equalent moment of inertia and equalent load torque referred to the motor shaft.

Ans:

Load 1 : Rotational motion.

$a = 0.1$

$\eta_1 = 90\% = 0.9$

$J_1 = 10\,kg\,m^2$.

$T_{l1} = 10\,Nm$.

Load 2 : Translational motion

$M = 1000\,kg$

$V = 1.5\,m/s$.

$\eta_2 = 85\% = 0.85$.

$F = M \times 9.81$

$F = 1000 \times 9.81 = 9810.N$

Motor

$J_0 = 0.2\,kg\,m^2$.

$N = 1420\,rpm$

$\omega_m = \frac{2\pi N}{60} = \frac{2\pi \times 1420}{60} = 148.7\ rad/sec$.

equalent moment of inertia, $J = J_0 + a^2 J_1 + M\left(\frac{V_1}{\omega_m}\right)^2$

$J = 0.2 + (0.1)^2 \times 10 + 1000 \times \left(\frac{1.5}{148.7}\right)^2 = 0.401\,kg\,m^2$.

equalent load torque

$T_l = \frac{a T_{l1}}{\eta_1} + \frac{F_1 V_1}{\eta_2 \omega_m}$

$= \frac{0.1 \times 10}{0.9} + \frac{(9810 \times 1.5)}{(0.85 \times 148.7)} = 117.53\,Nm$

Power developed by the motor = equalent torque $\times \omega_m$

$= 117.53 \times 148.7$

$= 17476.711\ W$

Components of load torque

Load torque T_L can be further divided into

1) Friction torque ~~Tf~~ T_F

Friction will be present at the motor shaft and also in the various parts of the load. T_F is the equivalent value of friction torques referred to the motor shaft.

2) Windage torque, T_W

When a motor runs, wind generates a torque opposing the motion. This is known as windage torque. It is proportional to the square of speed of the motor.

$$\boxed{T_W = C\omega_m^2}$$

where C is a constant.

3) Torque required to do the useful mechanical work, T_L.

Nature of this torque depends on particular application. It may be constant, some function or speed, or may depend on position or path followed by the load.

Friction torque T_F can be resolved into 3 components T_V, T_C and T_S.

a) Torque due to viscous friction, T_V

T_V is proportional to speed.

$$\therefore \boxed{T_V = B \times \omega_m}$$

where B – viscous friction coefficient.

b) Torque due to columb friction, T_C

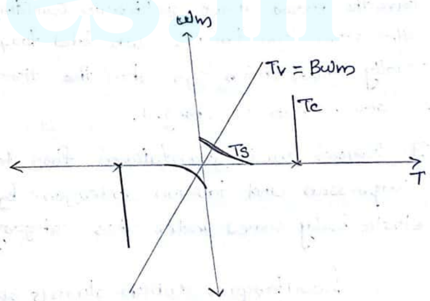
It is constant and independent of speed.

c) Torque due to static friction, T_S

T_S also known as stiction friction. Friction at zero speed is called stiction friction.

Inorder for drive to start it has to overcome static friction.

Speed – Torque characteristics



$$T_F = T_V + T_C + T_S$$

Therefore the motor torque eqn

$$T = T_l + J\frac{d\omega_m}{dt}$$

accelerating $T > T_l$

decelerating $T < T_l$

$$= T_F + T_W + T_L + J\frac{d\omega_m}{dt}$$

$$= T_V + T_C + T_S + T_W + T_L + J\frac{d\omega_m}{dt}$$

$$T = B\omega_m + T_C + T_S + C\omega_m^2 + T_L + J\frac{d\omega_m}{dt}$$

(imp) Classification of load torques

Various load torques can be broadly classified into 2 categories Active and passive load torques.

Load torques which have the potential to drive the motor under equilibrium conditions are called active load torques. Such load torque usually retain their sign when the direction of drive rotation is changed.

eg: Torques due to gravitational force, tension, compression and torsion undergone by an elastic body comes under this category.

Load torques which always opposes the motion and the changer change their sign on the reversal of motion are called Passive load torques.

Such torques are due to friction, cutting, etc.

Q. A drive has the following parameters.

$J = 10\,kgm^2$, $T = 100 - 0.1N$, Nm, Active load torque $T_l = 0.05N$, Nm. where N is the speed in rpm. Initially the drive is operating in steady state. Now it is to be reversed. For this motor characteristic is changed to $T = -100 - 0.1N$, Nm. Calculate the time of reversal.

Ans:

$$T = T_l + J\frac{d\omega_m}{dt} \longrightarrow \text{zero for steady state}$$

Initially the

For steady state speed $T - T_l = 0$.

$$100 - 0.1N - 0.05N = 0$$

$$100 - 0.15N = 0.$$

$$N = \frac{100}{0.15} = \cancel{2000}\ 666.6 \text{ rpm}$$

After reversal,

$$T = -100 - 0.1N, \text{ Nm}.$$

$T_l = 0.05N$ (Active load no sign changes)

$$T - T_l = 0$$

$$-100 - 0.1N - 0.05N = 0.$$

$$-100 - 0.15N = 0$$

$$N = -666.6 \text{ rpm}$$

Let $N_2 = \underline{-666.6}$ rpm

Considering the practical conditions $N_2 = 95\%$ of -6

$\therefore N_2 = -632.27$

$= \underline{\underline{-633.4}}$ rpm

$$\therefore T - T_l = J\frac{d\omega_m}{dt}$$

$$\frac{d\omega_m}{dt} = \frac{1}{J}(T - T_l)$$

$$= \frac{1}{\cancel{8}10}(-100 - 0.1 \times$$

$N = \frac{2\pi N}{60}$

$$\frac{dN}{dt} = \frac{60}{2\pi \times \cancel{10} J}(T - T_l)$$

$$= \frac{60}{2\pi \times 10}\left[\left(-100 - 0.1 \times \overset{N}{\cancel{-633.4}}\right) - 0.05 \times \overset{N}{\cancel{-633.4}}\right]$$

$$\cancel{\frac{dN}{dt} = -4.765}$$

$$\frac{dN}{dt} = \underline{\underline{-95.5 - 0.14325N}}$$

$$dt = \frac{dN}{-95.5 - 0.14325N}$$

$$\int dt = \int_{N_1}^{N_2} \frac{-dN}{95.5 + 0.14325N}$$

$$\boxed{\int \frac{dx}{ax+b} = \frac{1}{a}\log_e(ax+b)}$$

$$\therefore dt = -\int_{N_1}^{N_2} \frac{dN}{95.5 + 0.143N}.$$

$$= -\left[\frac{1}{0.143}\log_e(0.143N + 95.5)\right]_{666.67}^{-633.4}$$

$$= -\left[\frac{1}{0.143}\log_e(0.143 \times -633.4 + 95.5) - \frac{1}{0.143}\underset{ln}{\log_e}(0.143 \times 666.67 + 95.5)\right]$$

$$= -\left[\cancel{4.8412} - \cancel{15.94}\right] = \underline{\underline{\cancel{11}.0988}}$$

$$= -\left[\cancel{34.43} \right. \qquad = -\left[11.147 - 36.72\right] = \underline{\underline{25.57}}$$

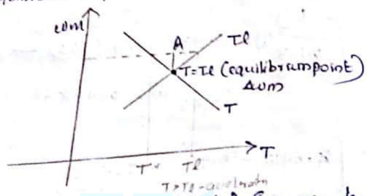
Steady state stability of an electrical drive

The drive is said to be in equilibrium if the torque developed by the motor is exactly equal to the load torque.

If the drive comes out of the state of equilibrium due to some disturbances, it comes back to the steady state for stable equilibrium but for unstable equilibrium the speed of drives increase uncontrollably or decrease to zero.

Steady state stability is defined as the capacity of the system which enable it to develop a force of such a nature as to restore equilibrium after any small disturbance.

Let us examine the steady state stability of equilibrium point A in Fig(1)



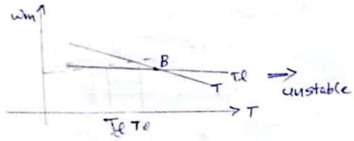
(stable) ($T_l > T$ - deceleration)

The equilibrium point A will be termed as stable when it can restored to it after a small departure from it due to a disturbance in the motor or load. Let the disturbance causes a reduction in speed be $\Delta\omega_m$.

At new speed, motor torque > load torque. Consequently motor will accelerate and operation will be restored to A.

Similarly, an increase in speed from equilibrium speed caused by a disturbance will make load torque > motor torque, resulting into deceleration and restoration of operation to point A. Hence the drive is steady state stable at point A.

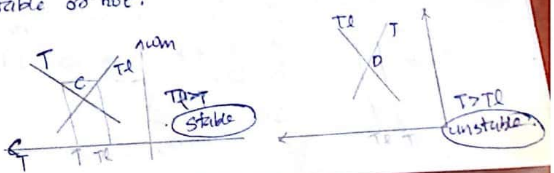
Let us now examine the equilibrium point B shown in Fig(2)



Point B is obtained when the same motor drives another load. A decrease in speed causes the load torque to become greater than the motor torque. The drive decelerates an operating point moves away from B.

Similarly, an increase in speed will make the motor torque > load torque which will move the operating point away from B. Thus B is an unstable point of equilibrium.

? Check the following equilibrium point of a drive is stable or not.



Derivation for steadystate stability criteria for a drive.

An equilibrium point will be stable when an increase in speed causes load torque to exceed the motor torque.

At equilibrium point,

$$\boxed{\frac{dT_l}{d\omega_m} > \frac{dT}{d\omega_m}}$$

It can be derived by an alternative approach. Let a small change in speed be $\Delta\omega_m$ which results in ΔT and ΔT_l changes in T and T_l respectively.

We know that,

$$T = T_l + J\frac{d\omega_m}{dt} \quad \text{---(1)}$$

$$T + \Delta T = (T_l + \Delta T_l) + J\frac{d(\omega_m + \Delta\omega_m)}{dt} \quad \text{---(2)}$$

(2) − (1)

$$\Delta T = \Delta T_l + J\cdot\frac{d\Delta\omega_m}{dt} \quad \text{---(3)}$$

For small changes, the speed−torque characteristic curves of motor and load can be assumed to be straight line. Therefore

$$\Delta T = \left(\frac{dT}{d\omega_m}\right)\omega_m$$

$$\Delta T_l = \left[\frac{dT_l}{d\omega_m}\right]\Delta\omega_m.$$

where $\frac{dT}{d\omega_m}$ and $\frac{dT_l}{d\omega_m}$ are respective slopes of the steady state speed-torque curves of motor and therefore eqn(3) become

$$\left(\frac{dT}{d\omega_m}\right)\Delta\omega_m = \left(\frac{dT_l}{d\omega_m}\right)\Delta\omega_m + J\cdot\frac{d\omega_m}{dT}.$$

$$J\cdot\frac{d\omega_m}{dt} + \left(\frac{dT_l}{d\omega_m}\right)\Delta\omega_m - \left[\frac{dT}{d\omega_m}\right]\Delta\omega_m = 0.$$

$$\frac{d\Delta\omega_m}{dt} + \frac{1}{J}\left(\frac{dT_l}{d\omega_m}\right)\Delta\omega_m - \left(\frac{dT}{d\omega_m}\right)\Delta\omega = 0.$$

This is a 1st order differential equation. If the initial deviation in speed at $t=0$, be $\Delta\omega_{m0}$, solution becomes

$$\Delta\omega_m = \Delta\omega_{m0}\exp\left(\frac{-1}{J}\left(\frac{dT_l}{d\omega_m} - \frac{dT}{d\omega_m}\right)\right)t.$$

An operating point will be stable when $\Delta\omega_m$ approach zero, as $t\to\infty$. For this, exponent must be negative. This yield the inequality.

$$\boxed{\frac{dT_l}{d\omega_m} > \frac{dT}{d\omega_m}}$$